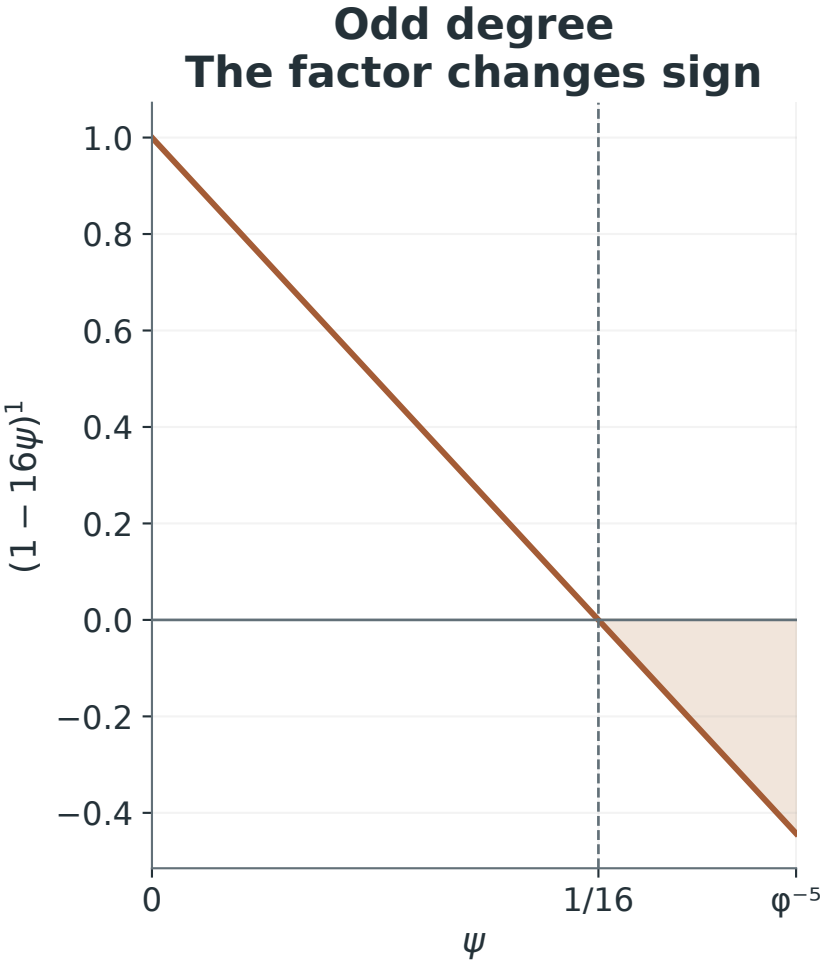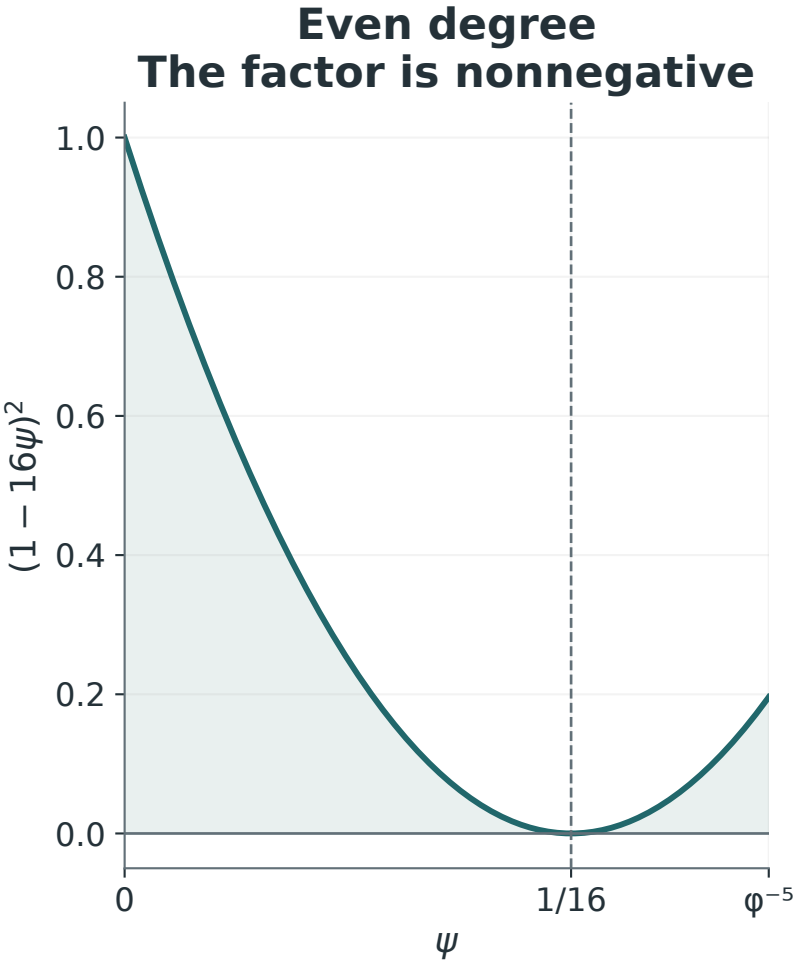$$\widetilde{J}_{n,m} = \int_{[0,1]^2} \psi^n (1 - 16\psi)^m \, d\mu$$